Naren Rachapalli

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Work Experience

Software Development Engineer, AI/ML Inference Jan 2025 – Present

Amazon Web Services Transcribe | Seattle, WA

Java, C++, AWS, Triton Inference Server, CUDA, PyTorch, Docker

- Improved GPU Automatic Speech Recognition (ASR) model inference throughput by ~4× using Triton Inference Server with FP32/BF16 mixed-precision quantization, dynamic batching, and shared CUDA memory optimization.
- Reduced streaming ASR word latency by ~333ms by redesigning chunk inference to use dynamic context windows with KV cache slicing, reducing dependency on future audio frames.
- Built an automated model evaluation platform that profiles inference throughput and latency regressions and benchmarks WER across spoken, written, and noisy datasets prior to launch.
- Built production GPU inference serving infrastructure with request validation, allowlisting, canary testing, alarms, and automatic CPU fallback to maintain high availability under real-time traffic.
- Increased peak traffic capacity by ~40% and reduced cold start latency by ~20% through predictive scaling, warm pools, and multi-instance-type support, contributing to service growth from ~$150M to ~$300M in ARR.

Software Development Engineer Sep 2024 – Dec 2024

Amazon Leo | Redmond, WA

Python, PyTest, Grafana, AWS, Docker

- Built a production testing framework to command and configure all satellite venues, enabling end-to-end validation across modulation schemes and simulated contact scenarios via GPS spoofing.
- Improved data processing performance by ~72% by creating Spark tables to summarize CQM and MAC data, automating legacy ETL workflows.
- Built real-time anomaly detection using Kolmogorov-Smirnov statistic and time-series forecasting via SageMaker on satellite telemetry streams.

Undergraduate Research Assistant Aug 2023 – May 2024

Purdue Computational Quantum Electromagnetics Lab | West Lafayette, IN

Python, NumPy, SciPy, Matplotlib

- Simulated multi-qubit, multi-cavity quantum system dynamics by generating configurable Hamiltonian matrices and solving Schrödinger's equation using numerical ODE methods.

Skills

Programming: Java, C/C++, Python, SQL, Shell Scripting
ML Infrastructure: Triton Inference Server, PyTorch, CUDA, TensorFlow
Infrastructure: AWS (EC2, SageMaker, ECS, CloudFormation), Docker, CI/CD, Spark, Git

Education

Purdue University West Lafayette, IN

B.S. Computer Science Dec 2024

Projects

Internal Knowledge MCP Server | *Python, Amazon Bedrock, MCP*

- Built an MCP server exposing internal ticketing, credential management, and account lookup as tools, backed by a Bedrock knowledge base over team documentation to accelerate debugging workflows.

Bitcoin Reinforcement Learning Trading Bot | *Python, TensorFlow, CUDA, NumPy, Pandas* | GitHub

- Built a CUDA-accelerated Deep Recurrent Q-Network trading bot achieving 74% ROI on historical Bitcoin data.

Email Extraction Automation Tool | *Python, NLTK, Scikit-Learn*

- Developed an NLP tool for eAlliance Corporation to extract order details from emails and integrated an RPA solution to automate SAP data retrieval and responses, reducing inquiry resolution time by 20%.